

Information Technology



Practical Assessment Task (PAT) Grade 12 – 2026

Task Guidelines

INTRODUCTION

Description

The Practical Assessment Task (PAT) is an assessment of a learner's individual interaction with the programming languages they have worked with, as well as with the various application packages they have encountered. The PAT is to be completed during the course of most of the Grade 12 year and should be formatively assessed so that learners have the opportunity to submit their best work. The information will finally be presented in several documents to communicate the coded solution in an integrated manner.

Purpose

The purpose of the PAT is to allow learners to demonstrate their skills in using the programming language that they have studied, i.e., Java or Delphi, to create an application that provides a solution to a specific problem.

This project will form 25% (100 marks) of the overall Grade 12 assessment.

Skills required

All skills acquired during the Grades 10 – 12 Information Technology curriculum will serve as guidelines to the minimum skills that should be demonstrated in the PAT.

Process

This project should allow for independent thought, work, and creative thinking. The PAT involves the learner choosing a scenario, finding and accessing information for the scenario, and developing a solution for that scenario. The solution must take into account managing and accessing stored data for the scenario. The PAT must be seen as a process that is gradually completed over a period of a few months.

Composition of the PAT

The PAT involves the following documents and a coded application. Every document must include a title page and a table of contents.

- ***Specifications Document***

This document details:

- a problem summary;
- the motivation for creating such a program accompanied by relevant research;
- the program's functions; and
- the data required for the program.

- ***Design Document***

This document is to detail the actual design of the program, namely:

- Program flow (how the program works – linked to the interface);
- User interface Description (what the screens should look like, theme, font, etc.);
- Secondary Storage design (how data is permanently stored and its structure);
- Explanation of Secondary Storage design;
- Class Diagrams for Backend Classes; and
- Explanation of which Backend Classes relate to secondary storage.

- ***Coding***

This requires an application to be developed based on the Specifications Document and Design Document, where emphasis must be placed on:

- secondary storage implementation;
- implementation of Backend classes;
- comments;
- user Interface implementation;
- separation of working code;
- good general programming techniques;
- defensive programming;
- fulfilment of specifications; and
- user experience.

- ***Technical & Testing Document***

This document outlines:

- externally sourced code;
- critical algorithms for the program;
- functional testing of the program;
- testing of two input variables; and
- the evaluation of the program.

Review and monitoring

As this project has a large scope, much of the work will have to be done outside of class time. The entire process will be facilitated and monitored by the teacher, who will provide timeframes for checking and submitting the various aspects. Continuous checking and feedback at regular intervals will ensure that learners obtain the best possible mark.

Educators will assess the Program Code, the Technical aspects and the Testing aspects in conjunction with an interview with the learner.

Resources

Learners are encouraged to use subroutines / classes / units from other programmers (duly acknowledged) to improve the quality of their program. When Generative AI code is used, learners must acknowledge and explain their usage of generative AI in terms of content, process, and resulting product.

It is the learner's responsibility to ensure that all borrowed, or AI-generated code is properly referenced (preferably using the Harvard standard) and acknowledged.

Borrowed or generated code may be used for this task up to a **maximum of 20 percent** when compared to the learner's own code.

Educators will interview learners to establish and ascertain how much borrowed code is in the project.

Assessment Rubric

A detailed assessment rubric is provided to facilitate marking, internal, and external moderation of the PAT. The rubric stipulates the evidence required for each aspect of the PAT.

Learners are encouraged to study the assessment rubric in order to fulfil the requirements of each aspect as well as help them improve on the subsequent aspects.

The four aspects should be assessed as follows:

Aspect	Marks	Recommended class time spent	Recommended submission time frame
Specifications Document	15	1 week	February
Design Document	30	2 weeks	March
Coding	40	6 weeks	July
Technical & Testing Document	15	1 week	August
TOTAL	100	12 weeks	

Topic and Content

Learners should choose any topic that allows for independent and creative thinking, and may be a solution to a real-world problem, a business management system, or even a game.

The topic can be implemented on one or more desktop computers, as a mobile application, or on a separate hardware device such as Raspberry PI, Arduino, or any combination thereof.

Educators must ensure that learners have written a substantial amount of their solution and that they have not relied too heavily on utilities and functions provided by a solution development tool or Generative AI.

To reduce the risk of plagiarising an existing application, many of which are available on the Internet, the learner must **code a completely new solution for the topic and scenario**.

Learners are expected to research existing solutions to their topic to determine whether there is an existing solution. They will also have to indicate how their project will differ from existing solutions.

The following types of projects will be accepted:

- projects involving the use of multiple database tables/text files;
- projects involving the use of data collected via the use of sensors; and
- games or gaming engines.

All projects will be accepted on the premise that the following are strictly adhered to:

- A considerable amount of meaningful data **MUST** be stored on a secondary storage device.
- Significant use of Backend classes is evident where the bulk of the processing occurs.

Planning and Submission

It is vital to start this project early in the year and that learners are given timeframes for when aspects are due.

Educators should check learners' topics early and advise on its scope and functionality. This must be done to ensure that learners are on the right path from the outset.

Educators may want to dedicate a lesson in the timetable for PAT to check on learners' progress and to give advice periodically. Learners must also address any changes in previously marked/assessed work.

The following planning/checklists are to be used **in conjunction with the PAT Assessment Rubric** to ensure that requirements are met.

SPECIFICATIONS DOCUMENT**Learner Name:** _____ **Due Date:** _____**Refer to the Assessment Rubric.**

In completing this document, the learners will have to:

- provide a problem summary, including the purpose of the program, a summary of functions, and a description of the target user group(s). Learners must also expand on it to describe exactly what needs to be done in their own words;
- conduct research to determine if there are existing software programs and provide an explanation of how the proposed project will differ from existing programs;
- list exactly what the functions of the program are going to be; and
- list all data required.

Submission of evidence

Hand in a **Specifications Document** prepared in an appropriate application, which contains the following:

Aspect	✓/✗	Comment
Problem Summary		
Research and Motivation		
Program Functions		
Data required		

DESIGN DOCUMENT**Learner Name:** _____ **Due Date:** _____**Refer to the Assessment Rubric.**

In completing this document, the learners will have to:

- provide a flow diagram of how the program works.
- provide a User Interface description.
- provide a Secondary Storage Design.
- provide an explanation for Secondary Storage Design.
- provide Class Diagrams for Backend classes.
- provide an explanation for how Backend classes relate to Secondary Storage.

Submission of evidenceHand in a **Design Document** prepared in an appropriate application, which contains the following:

Aspect	✓/✗	Comment
Program Flow		
User Interface Description		
Secondary Storage Design		
Explanation of Secondary Storage Design		
Class Diagrams		
Explanation of how Backend classes relate to Secondary Storage		

CODING**Learner Name:** _____ **Due Date:** _____**Refer to the Assessment Rubric.**

In completing this task, the learners will have to write the actual code for the intended program and:

- ensure secondary storage implementation;
- ensure implementation of Backend classes;
- provide comments in code;
- ensure the implementation of the User Interface;
- ensure separation of UI from Working Code;
- ensure inter-code communication between classes and within a class;
- show good general programming techniques;
- illustrate defensive programming techniques;
- ensure the fulfilment of program functions, as specified;
- provide an overall excellent user experience; and
- provide all **electronic files** used for processing/manipulating the data and running the program.

Submission of evidence (Teacher to interview learners)

Hand in the entire program containing the **code** and necessary files for the program to run.

Aspect	✓/✗	Comment
Secondary Storage Implementation		
Implementation of Backend Classes		
Comments		
User Interface Implementation		
Separation of Working Code		
Good General Programming Techniques		
Defensive Programming		
Fulfilment of Specifications		
User Experience		

TECHNICAL & TESTING DOCUMENT**Learner Name:** _____ **Due Date:** _____**Please refer to the Assessment Rubric.**

In completing this document, the learners will have to:

- list and reference all externally sourced code;
- provide pseudocode and explanations of algorithms vital to the functioning of the program;
- provide TWO sets of Functional testing;
- provide a Test Plan and Results for TWO input variables; and
- provide an evaluation of the program.

Submission of evidence (Teacher to interview learners)Hand in a **Technical and Testing Document** prepared in an appropriate application, which contains the following:

Aspect	☐ / ☐	Comment
Externally Sourced Code		
Critical Algorithms		
Functional Testing		
Test Plan and Results for TWO Input Variables		
Evaluation of the Program		



INFORMATION TECHNOLOGY PRACTICAL ASSESSMENT TASK DECLARATION

Learner Exam Number	
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DECLARATION BY THE LEARNER:

I, _____ (learner full name)
hereby confirm that:

- I have read and understood my school's policy relating to Academic Dishonesty.
- I have adhered to my school's policy regarding Academic Dishonesty.
- The PAT submitted has been completed in full by me.
- I have clearly indicated in my PAT any sections of work/code that were generated or assisted by AI or where code was borrowed, giving credit where it is due.
- I acknowledge that I may have to verbally explain my work presented in this PAT.
- I further declare that my PAT code comprises 20% or less code that is borrowed or from Generative AI, as required by the IEB, and this is clearly indicated in my documentation.

Signed: _____ Date: _____
Learner

DECLARATION BY TEACHER:

I, _____ (teacher full name & title)
at _____ (name of school)

declare that the work provided by this learner has been monitored and checked for plagiarism.

Signed: _____ Date: _____
Teacher